IIBS Computer Lab Management System

Complete Step-by-Step Installation, Lockdown & Deployment Guide

Prerequisites Checklist

Item	Requirements
Teacher Master PC	Windows 10/11, Python 3.8+ installed, connected to Lab LAN Switch/Router.
Student Lab PCs	Windows 10/11 connected to the same LAN switch/subnet.
Network	Local Area Network (LAN) / Ethernet Switch. External Internet is NOT required.

PART 1: Setup Master Server (Teacher's PC)

Step 1.1: Determine Teacher PC's LAN IP Address

1. Press **Win + R**, type cmd, and press Enter.
2. Run the command: ipconfig
3. Note down your **IPv4 Address** (e.g., 192.168.1.100). Student PCs will connect to this address.

Step 1.2: Launch the Master Server

1. Navigate to d:\Antigravity\IIBS\lab_manager\ on the teacher's PC.
2. Double-click **start_server.bat**.
3. The batch script will automatically ensure dependencies (Flask, Werkzeug) are installed and start the server on port 5000.

Step 1.3: Access the Web Control Portal

Open any web browser (Chrome, Edge) and visit:

```
http://127.0.0.1:5000/admin (or from any device: http://<TEACHER_IP>:5000/admin)
```

PART 2: Setup Student Lab PCs

Method A: Standalone Executable (.exe) — Recommended (No Python Needed on Students)

1. On the Teacher/Development PC, open Command Prompt and compile the client into an executable:

```
pip install pyinstaller cd /d d:\Antigravity\IIBS\lab_manager\client pyinstaller --onefile
--noconsole client.py
```

2. Inside lab_manager\client\dist\, you will find **client.exe**.
3. Copy two files to a USB drive: **client.exe** and **client_config.json**.
4. On each student PC, create folder C:\LabClient\ and paste both files there.
5. Open client_config.json in Notepad and set the Teacher's LAN IP address:

```
{ "server_url": "http://192.168.1.100:5000" }
```

Method B: Direct Script Execution (If Python is Already Installed)

Copy the lab_manager\client\ folder to the student PC, configure the IP in client_config.json, and double-click **start_client.bat**.

PART 3: Student PC Security & Non-Admin Lockdown

Step 3.1: Create Standard (Non-Admin) Account

1. Open Windows **Settings > Accounts > Other users**.
2. Click **Add account** and create a local user named **Student**.
3. Ensure the Account Type remains **Standard User** (do NOT grant Administrator rights).

Step 3.2: Configure Auto-Start on Boot

1. Log in to the Student account.
2. Press **Win + R**, type shell:startup, and press Enter.
3. Create a shortcut to C:\LabClient\client.exe (or start_client.bat) inside this folder.

Result: Whenever the student machine boots, the kiosk lock screen launches automatically.

Step 3.3: Disable Task Manager via Group Policy (Prevents Bypassing)

1. Press **Win + R**, type gpedit.msc, and press Enter.
2. Navigate to: **User Configuration > Administrative Templates > System > Ctrl+Alt+Del Options**.
3. Double-click **Remove Task Manager**, select **Enabled**, and click OK.

Result: Students cannot open Task Manager to kill the client process.

PART 4: Daily Operations & Verification Guide

Operation	How It Works & Admin Procedure
Student Registration	On the kiosk lock screen, the student clicks **New Student Register**, enters Roll No, Full Name, Username, and Password. Account is saved on the Master Server, allowing instant login.
1-Click Exam Mode (Internet Cutoff)	In the Admin Portal, click **'Enable Exam Mode (Block Internet)'**. All student PCs switch to normal usable desktop mode, but external internet (ports 80, 443, 53) is blocked via Windows Firewall. Click **'End Exam & Restore Internet'** when finished.
One-Shot File Broadcast	In Admin Portal > *File Broadcast*, upload any assignment or PDF. All logged-in student PCs download it automatically to Desktop\Lab_Shared_Files\.
1-Hour Idle Auto-Shutdown	Tracks native Windows mouse/keyboard movement. If inactive for 60 minutes, the client notifies the Master Server and executes shutdown /s /t 30.
Password Reset	Search any student in the Admin Portal, click **Reset Password**, enter a new password, and click Save. The student can log in immediately.
Audit Logs Export	Click **'Export Logs (CSV)'** on the portal to download a spreadsheet report of all registrations, logins, exam sessions, and shutdowns.